

# Curriculum Vitae

## Roni Gattegno



### Personal details

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I.D.: 315370825

Languages: Hebrew, English, Spanish

### Education

2021-2026: Direct PhD student in Neuroscience, Sagol School of Neuroscience, Tel Aviv University, Noninvasive therapy and diagnostic of glioblastoma using ultrasound and nanobubbles.

2021-2023: MSc in Neuroscience, Sagol School of Neuroscience, Tel Aviv University, Prof. Tali Ilovitsh

2017-2021: BSc in Biomedical Engineering, The Iby and Aladar Fleischman Faculty of Engineering, Tel Aviv University. Magna cum Laude.

2017-2021: BSc in Biology with emphasis in Neuroscience, Sagol School of Neuroscience, Tel Aviv University. Magna cum Laude.

### Professional Experience and Appointments

2025: Visiting researcher at Konofagou lab, Columbia University, New York, June-August 2025.

2022: Visiting researcher through Young Investigator Training Program (YITP) at Barral lab, Institut de l'Audition, Institut Pasteur, Paris, July 2022.

2018-2022: Research assistant at the laboratory of Dr. Eran Stark, Sagol School of Neuroscience, Tel Aviv University.

### Teaching

2023-2026: Teaching assistant, Radiation and Imaging in Medicine, Biomedical Engineering Department, Tel Aviv University.

2023-2026: Teaching assistant, Laboratory in Biomedical Engineering, Biomedical Engineering Department, Tel Aviv University.

2023-2026: Teaching assistant, Biological Control Systems, Biomedical Engineering Department, Tel Aviv University.

2022: Teaching assistant, Mathematical Modeling of Neurons and Networks, Sagol School of Neuroscience, Tel Aviv University.

2020-2023: Teaching assistant, Python, Extra Student.

2020-2023: Teaching assistant, Partial Differential Equations, Extra Student.

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## Military Service

2015-2017: In Unit 8200, The IDF's Technological Intelligence unit.

- Graduated from "MATAM" course, as a "COMINT" operator.
- Title of excellence in the course.
- Second position as a product manager in the unit.

## Honors and Awards (including Membership in Professional Societies):

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|-----------|---------------------------------------------------------------------------------|
| 2025      | FENS regional meeting travel grant.                                             |
| 2024      | The BSF Prof. Rahamimoff travel grants for young scientists.                    |
| 2024      | Israel academy Adams fellowships program.                                       |
| 2024      | The Cancer Biology Research Center Travel Award.                                |
| 2024      | The Marian Gertner Institute for Medical Nanosystems - Excellence Student Award |
| 2023-2025 | IEEE membership.                                                                |
| 2023      | Teva BioInnovators student excellence fellowship.                               |
| 2023      | Student award poster presentation, ISTU.                                        |
| 2022      | FENS travel grant.                                                              |
| 2022      | Sagol School of Neuroscience dean's honor list.                                 |
| 2021      | Sagol School of Neuroscience dean's honor list.                                 |
| 2019      | Sagol School of Neuroscience dean's honor list.                                 |
| 2021      | "Hamerkaz Haezrahi" scholarship.                                                |
| 2020      | Salti foundation scholarship.                                                   |
| 2018-2020 | Excellence program of the faculty of engineering.                               |

## Published Publications:

- 1) Elbaz M, Ad-El N, Chulanova Y, Brier D, Goldsmith M, Bismuth M, Brosque A, **Gattegno R**, Sher D, Gutkin A, Bar-On D, Friedmann-Morvinski D, Peer D, Ilovitsh T (2025) "Low-frequency ultrasound-mediated blood-brain barrier opening enables non-invasive lipid nanoparticle RNA delivery to glioblastoma". *Journal of Controlled Release*, 114018. **(Impact factor 11.46, Q1)**
  - 2) Glickstein B, Bismuth M, **Gattegno R**, Bercovici T, Shaul O, Aronovich R, Horn G, Levin AG, Feng Y, Ilovitsh T (2025) "Volumetric Nanodroplet-Enhanced Ultrasound Surgery Combined with Immune Checkpoint Inhibition as a Cancer Therapy Platform". *Small*, 21(23), p.2411474. **(Impact factor 13, Q1)**
  - 3) **Gattegno R**, Arbel L, Reiss N, Katz S, Ilovitsh T (2023) "Enhanced capillary delivery with nanobubble-mediated blood-brain barrier opening and advanced high resolution vascular segmentation". *Journal of Controlled Release* 369, 506-516. **(Impact factor 11.46, Q1)**
  - 4) Peko L, Katz S, **Gattegno R**, Ilovitsh T (2023) "Protocol to assess extravasation of fluorescent molecules in mice after ultrasound-mediated blood-brain barrier opening". In press, *STAR Protocols*. **(Impact factor 1.3, Q1)**
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- 5) Katz S, **Gattegno R**, Peko L, Zarik R, Hagani Y, Ilovitsh T (2023) "Diameter-dependent assessment of microvascular leakage following ultrasound-mediated blood-brain barrier opening". *iScience* 26 (6). **(Impact factor 4.1, Q1)**
- 6) Sloin HE, Spivak L, Levi A, **Gattegno R**, Someck S, Stark E (2023) "Local activation of CA1 pyramidal cells induces theta phase precession". *Science*, 383(6682), 551-558. **(Impact factor 45.8, Q1)**
- 7) Someck S, Levi A, Sloin HE, Spivak L, **Gattegno R**, Stark E (2023) "Positive and biphasic extracellular waveforms correspond to return currents and axonal spikes". *Communications Biology*, 6:950. **(Impact factor 6.5, Q1)**
- 8) Sloin HE\*, Bikovski L\*, Levi A, Amber-Vitos O, Katz T, Spivak L, Someck S, **Gattegno R**, Sivroni S, Sjulson L, Stark E (2022) "Hybrid offspring of C57BL/6J mice exhibit improved properties for neurobehavioral research". *eNeuro*, 9:ENEURO. 0221-22.2022. **(Impact factor 2.7, Q1)**

## Other Noteworthy Activities and Skills

### a. Conferences

- 2025: Oral presentation, "Noninvasive biomarker discovery via sonobiopsy and proteomics after BBB opening in a Glioblastoma model", IUS, September 2025.
- 2025: Poster acceptance, "Nanobubble-mediated sonobiopsy for enhanced protein biomarker detection in glioblastoma", FENS regional meeting, June 2025.
- 2024: Oral presentation + best poster competition finalist, "Enhanced blood-brain barrier opening with nanobubble-mediated focused ultrasound and deep learning vascular segmentation", ISTU, September 2024.
- 2024: Oral presentation, "Enhanced capillary delivery with nanobubble-mediated blood-brain barrier disruption and advanced high resolution vascular segmentation", IUS, September 2024.
- 2023: Oral presentation + best poster competition finalist, "Effect of microvasculature diameter on FUS-mediated BBB opening", ISTU, April 2023.
- 2022: Poster presentation, "Extracellular recordings of multiple optically-tagged neurons for cell type classification", FENS forum, July 2022.

### b. Extracurricular

- 2025: Yazamiyot on Campus Program, Entrepreneurship Center, Tel Aviv University. Won 2<sup>nd</sup> place with Digivet project.
- 2021-2023: Engineers Without Borders, Led the Assistive Technology team, developing accessibility projects across Israel.
- 2021: "Hevruta" - a program that aims to bridge societal gaps between non-religious and ultra-orthodox women.
- 2019-2020: "Mithabrim+" by the dean of students - participated in a course on nanosatellites and taught science to young girls from the social periphery.
- 2018-2020: Tutor in "Walking Together" - a mentoring program for first-year bachelor's students.
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